5) Experiment to detect outliers in a given data set. Description: Understand the procedure to identify the outliers in a given dataset

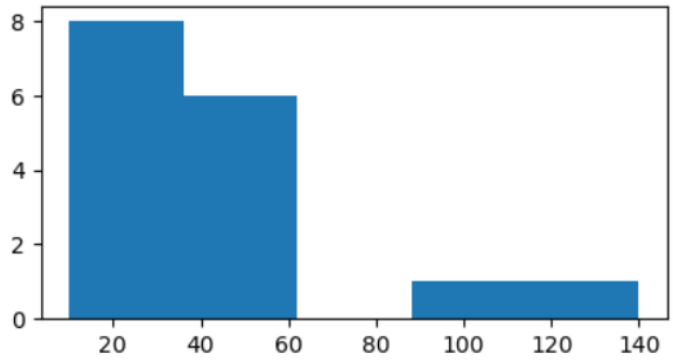
Program:

```
import numpy as np
import matplotlib.pyplot as plt
array=np.array([10,15,19,22,25,28,31,34,37,39,42,44,47,49,110,140])
print("Array:")
print(array)
print("Mean:")
print(array.mean())
print("25th Percentile:")
print(np.percentile(array,25))
print("50th Percentile:")
print(np.percentile(array,50))
print("75th Percentile:")
print(np.percentile(array,75))
print("100th Percentile:")
print(np.percentile(array,100))
def outDetection(array):
    Q1,Q3=np.percentile(array,[25,75])
    IQR=Q3-Q1
    lr=Q1-(1.5*IQR)
    ur=Q3+(1.5*IQR)
    return lr,ur
lr,ur=outDetection(array)
print("Lower Range:")
print(lr)
print("Upper Range:")
```

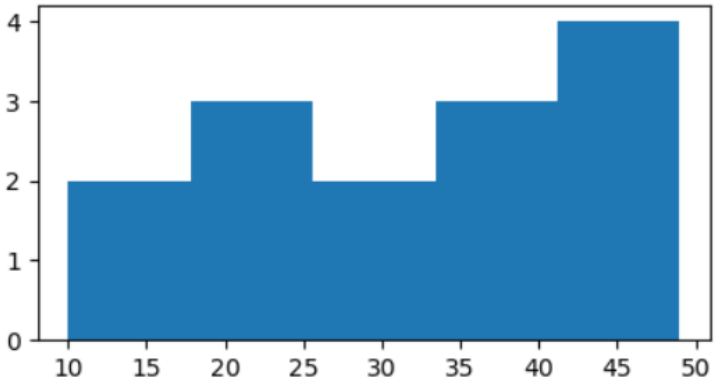
```
print(ur)
plt.figure(figsize=(5,2.5))
plt.hist(array,bins=5)
plt.show()
new_array=array[(array>lr)&(array<ur)]
print("Array After Removing Outliers:")
print(new_array)
plt.figure(figsize=(5,2.5))
plt.hist(new_array,bins=5)
plt.show()
lr1,ur1=outDetection(new_array)
print("New Lower Range:")
print(lr1)
print("New Upper Range:")
print(ur1)
final_array=new_array[(new_array>lr1)&(new_array<ur1)]
print("Final Array:")
print(final_array)
plt.figure(figsize=(5,2.5))
plt.hist(final_array,bins=5)
plt.show()
```

Output:

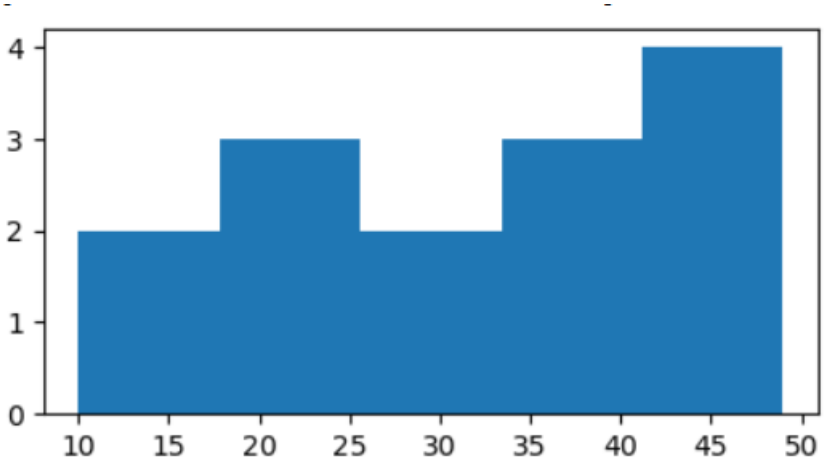
```
Array:
[ 10  15  19  22  25  28  31  34  37  39  42  44  47  49 110 140]
Mean:
43.25
25th Percentile:
24.25
50th Percentile:
35.5
75th Percentile:
44.75
100th Percentile:
140.0
Lower Range:
-6.5
Upper Range:
75.5
```




```
Array After Removing Outliers:
[10 15 19 22 25 28 31 34 37 39 42 44 47 49]
```




```
New Lower Range:
-5.0
New Upper Range:
69.0
Final Array:
[10 15 19 22 25 28 31 34 37 39 42 44 47 49]
```



Result:

Experiments to detect the outliers for the data is conducted.